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Experiment-4

Aim: To set up a Continuous Integration Pipeline using Jenkins Act

Lab Outcome: Mapped LO3

LO3: Build and manage automated CI/CD pipelines for application development, testing, packaging and deployment.

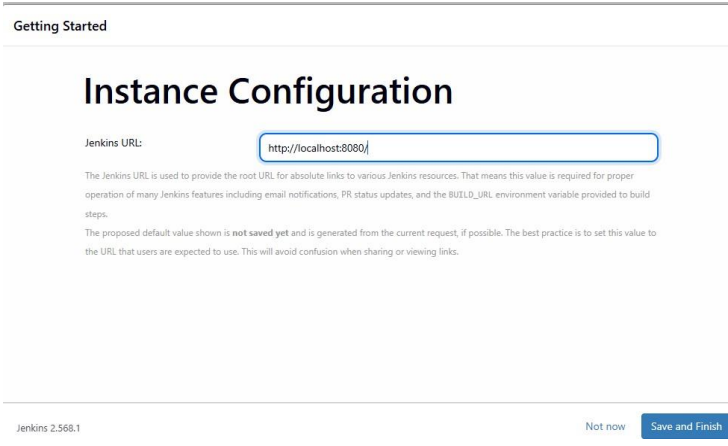
Java Installation:

```
C:\Users\Lenovo>java -version
openjdk version "21.0.12" 2026-07-21 LTS
OpenJDK Runtime Environment Microsoft-14653896 (build 21.0.12+8-LTS)
OpenJDK 64-Bit Server VM Microsoft-14653896 (build 21.0.12+8-LTS, mixed mode, sharing)
```

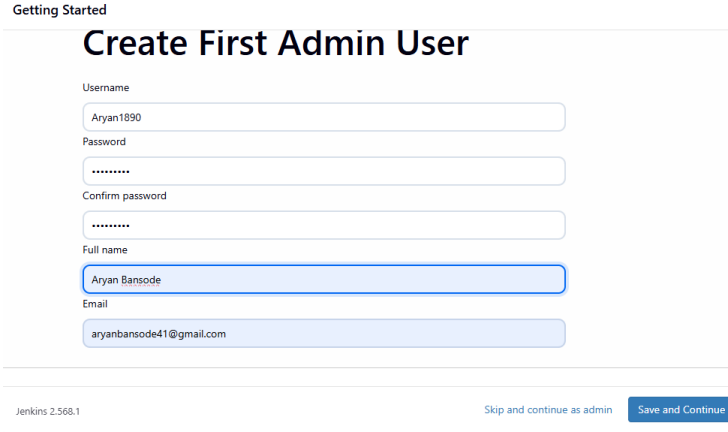
Navigate To Directory:

```
[CRLF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[CRLF]> Please use the following password to proceed to installation:
[CRLF]>
[CRLF]> 6d871f0fc0f149ef94a4a1a14fd6f424
[CRLF]>
[CRLF]> This may also be found at: C:\Users\Lenovo\.jenkins\secrets\initialAdminPassword
[CRLF]>
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]> ************************************************************
```

Instance Configuration:



To create First Admin User


Getting Started
Create First Admin User
Username
Aryan1890
Password
Confirm password
Full name
Aryan Bansode
Email
aryanbansode41@gmail.com
Jenkins 2.568.1
Skip and continue as admin
Save and Continue

S

Dashboard:


Jenkins
Aryan Bansode
Aryan Bansode
Add description
Profile
Username: Aryan1890
Builds
My Views
Account
Appearance
Preferences
Security
Experiments
Credentials
REST API
Jenkins 2.568.1

AIM: To build pipelineusing Jenkins.

THEORY:

What is a Pipeline in Jenkins?
A Jenkins Pipeline is a set of automated processes that help developers and DevOps teams build, test, and deploy code in a continuous integration/continuous delivery (CI/CD) environment. In simple terms, it's a workflow that defines how software moves from code to production in an automated and repeatable manner.

Key Benefits of Jenkins Pipeline:

- Automates CI/CD processes

- Provides better visibility into the build process
- Offers version control of build logic (Pipeline as Code)
- Supports complex workflows (parallel execution, conditional logic, etc.)

Types of Jenkins Pipelines:

There are two main types of syntax used to define Jenkins pipelines:

1. Declarative Pipeline

Simpler and more structured syntax Designed for users who want an easy-to-read format Introduced later to make pipelines more readable and maintainable

Key Features:

- Uses pipeline {} block

- Organized into stages and steps
- Easier error handling
- Better validation in Jenkins UI

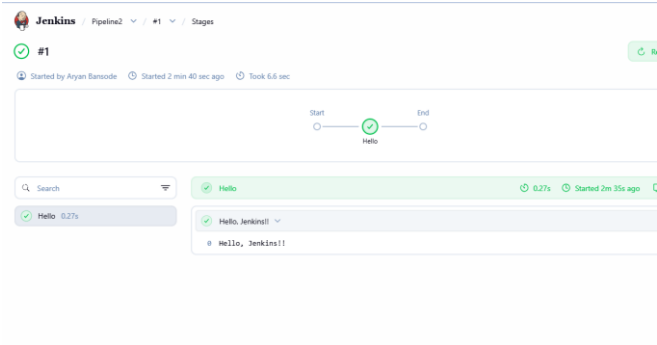
2. Scripted Pipeline

- More flexible and powerful
- Written in Groovy-based syntax
- Gives full control over the build process
- Suitable for advanced users

Key Features:

- Uses node {} block
- Everything is written as code
- Allows complex scripting and logic
- Less user-friendly for beginn

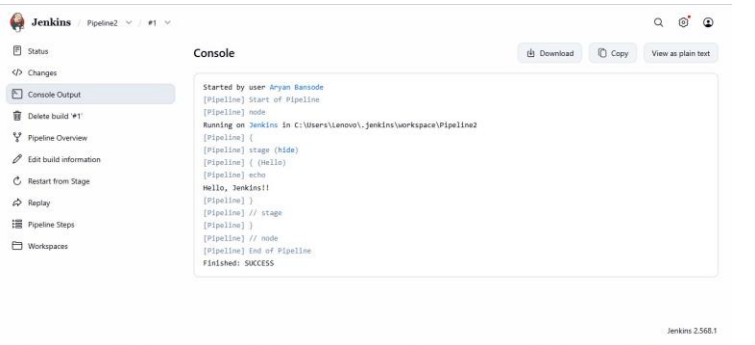
Creation:



Code:

```
pipeline {
    agent any
    stages {
        stage('Hello') {
            steps {
                echo 'Hello, Jenkins!!'
            }
        }
    }
}
```

Output:



CONCLUSION: LO1, LO3

We learned how to create Jenkins Pipelines to automate the build, test, and deployment process. We explored both Declarative and Scripted pipelines, understanding their structure, benefits, and use cases. By implementing different pipeline examples, we demonstrated how Jenkins enables efficient, repeatable, and reliable CI/CD workflows.